

# ◆ PROJECT SYNOPSIS ◆



## PREGNANCY COMPLICATION PREDICTION USING MACHINE LEARNING

Early Prediction • Smarter Monitoring • Healthier Future

|            |         |                   |
|------------|---------|-------------------|
| PCP        | 500     | 86%               |
| ML PROJECT | RECORDS | REPORTED ACCURACY |

### 1. Project & Team Details

| Name                  | College | Branch | Mobile                                                        | Email                    | Role   |
|-----------------------|---------|--------|---------------------------------------------------------------|--------------------------|--------|
| Ahir Diya Bharatbhai  | VTBCSR  | BCA    | 7434093194                                                    | ahirdiya977@gmail.com    | Lead   |
| Dhole Neha Bharatbhai | VTBCSR  | BCA    | 8511114033                                                    | bed143243@gmail.com      | Member |
| Dhimmar Mahi Anilbhai | VTBCSR  | BCA    | 9558216280                                                    | mahidhimmar038@gmail.com | Member |
| Program Track         |         |        | Domain (Analytics / Machine Learning / Deep Learning / GenAI) |                          |        |
| Science-Spoke         |         |        | Machine Learning                                              |                          |        |

### Guide / Mentor

Prajapati Jyoti Harinarayanbhai

### 2. Problem Statement

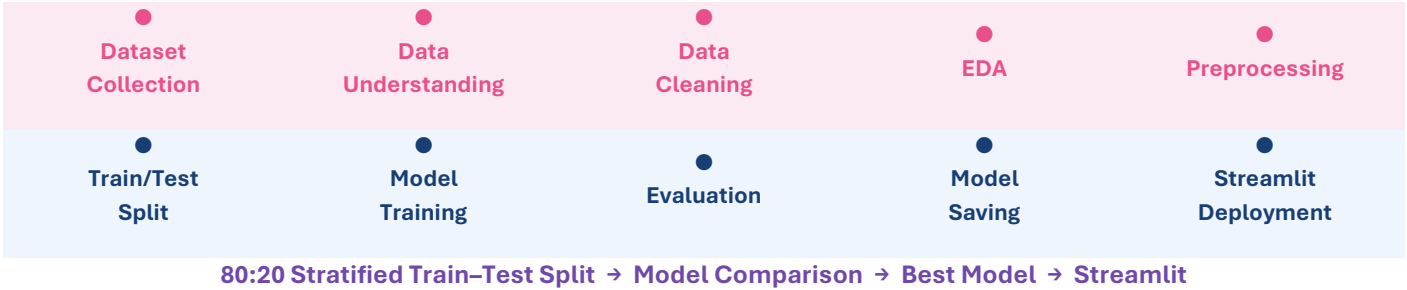
#### The Challenge

Pregnancy complications can create significant risks for both the mother and baby. This project develops a machine learning classification system to identify whether a pregnancy record indicates a complication or no complication, using clinical and pregnancy-related attributes. A Streamlit interface will provide a convenient preliminary prediction tool.

### 3. Key Objectives

- Understand and analyze clinical and pregnancy-related data.
- Perform data cleaning, exploratory data analysis (EDA) and preprocessing.
- Develop machine learning classification models for pregnancy complication prediction.
- Implement and compare Logistic Regression, Decision Tree, Random Forest and XGBoost.
- Evaluate models using Accuracy, Precision, Recall, F1-Score and ROC-AUC.
- Identify an effective predictive model and build a user-friendly Streamlit application.

## 4. Methodology / Implementation Plan



## Algorithms / Models

|                                                                                                       |                                                                                                 |                                                                                                 |                                                                                             |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
|  Logistic Regression |  Decision Tree |  Random Forest |  XGBoost |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|

## Evaluation & Analysis

- Accuracy — overall classification correctness.
- Precision — correctness of positive predictions.
- Recall — ability to identify complication cases.
- F1-Score — balance between Precision and Recall.
- ROC-AUC — model discrimination performance.
- Confusion Matrix and ROC Curve — visual model analysis.

## Tools / Technologies

|                      |                  |           |         |              |
|----------------------|------------------|-----------|---------|--------------|
| Python               | Jupyter Notebook | Pandas    | NumPy   | Scikit-learn |
| Matplotlib / Seaborn | Joblib           | Streamlit | VS Code |              |

## Dataset Information

|         |                                                            |
|---------|------------------------------------------------------------|
| Records | 500                                                        |
| Columns | 17 (16 input features + 1 target)                          |
| Target  | Pregnancy_Complication                                     |
| Classes | 0 = No Pregnancy Complication   1 = Pregnancy Complication |

## 5. Resources & Project Timeline

### Hardware & Software Requirements

- Laptop / PC with sufficient RAM and storage.
- Operating System: Windows / Linux / macOS.
- Programming Language: Python.
- Development Tools: Jupyter Notebook and VS Code.
- Libraries / Frameworks: Pandas, NumPy, Scikit-learn, Matplotlib/Seaborn and Streamlit.

### Project Timeline

| Phase   | Task                                      | Completion |
|---------|-------------------------------------------|------------|
| Phase 1 | Dataset collection and understanding      | Week 1     |
| Phase 2 | Data cleaning, EDA and visualization      | Week 2     |
| Phase 3 | Preprocessing and train-test split        | Week 3     |
| Phase 4 | Model development and comparison          | Week 4     |
| Phase 5 | Performance evaluation and interpretation | Week 5     |
| Phase 6 | Model saving and Streamlit deployment     | Week 6     |
| Phase 7 | Testing, documentation and final demo     | Week 7     |

## 6. Expected Outcomes / Conclusion

### Expected Result

A functional machine learning system capable of classifying pregnancy records into no pregnancy complication or pregnancy complication classes, with comparison of multiple classification algorithms and standard evaluation metrics.

### Observed Run (as reported in the source synopsis)

Optimized XGBoost model achieved 86% accuracy and 0.9531 ROC-AUC in the observed run.

### Application

The trained model will be saved and integrated into a Streamlit-based application where users can enter clinical and pregnancy-related parameters and view the predicted risk class.

### Important Note

This system is intended as a machine-learning-based preliminary prediction tool and is not a replacement for professional medical diagnosis.

## 7. References

- Project dataset and accompanying project materials provided for the Pregnancy Complication Prediction project.
- Scikit-learn documentation for machine learning classification and model evaluation.
- Python documentation and Jupyter Notebook resources.
- Streamlit documentation for application deployment.